RECRUITMENT & PLACEMENT MANAGEMENT SYSTEM

Python + MySQL Project Abstract

Project Abstract

The **Recruitment & Placement Management System** is a Python and MySQL based application designed to automate and manage the recruitment and placement activities of students or candidates in a training institute or placement organization.

The system provides centralized management of candidates, companies, job requirements, job applications, eligibility validation, interview rounds, feedback, offers, joining details, and final placements.

The application is developed using **Python, Object-Oriented Programming, MySQL connectivity, Collections/Data Structures, Exception Handling, File Handling, Modules, Functions, and Database Programming**. The project follows a modular architecture in which models, database operations, business services, validations, utilities, reports, and application flow are separated into different modules.

A major component of the application is the **Candidate Eligibility Engine**. It automatically checks job requirements such as qualification, percentage, passing year, technology, skills, backlog criteria, and job status before allowing a candidate to apply.

The complete recruitment workflow is managed through defined business stages: **Candidate Registration -> Eligibility Verification -> Application -> Shortlisting -> Interview Scheduling -> Interview Feedback -> Selection -> Offer -> Joining -> Placement.**

The system also implements industry-oriented requirements such as duplicate application prevention, candidate and job status management, interview round tracking, placement history, business validations, custom exception handling, logging, search, filtering, and management reports.

MySQL is used as the backend database for maintaining relational data and generating reports using joins, grouping, filtering, and aggregate queries. Python database connectivity is used to perform secure CRUD and transactional operations.

HTML and CSS pages are created only as **UI prototypes** for login, dashboard, candidates, jobs, applications, interviews, offers, placements, and reports. These prototype pages are not connected to the Python backend and represent the proposed future user interface.

The project provides practical exposure to Python programming, database connectivity, OOP, modular programming, business logic implementation, SQL, exception handling, reporting, and real-world recruitment and placement workflows.

Project Type: Industry-oriented backend business application with HTML/CSS UI prototype.